

Decision Under Uncertainty

States, Actions and Policies

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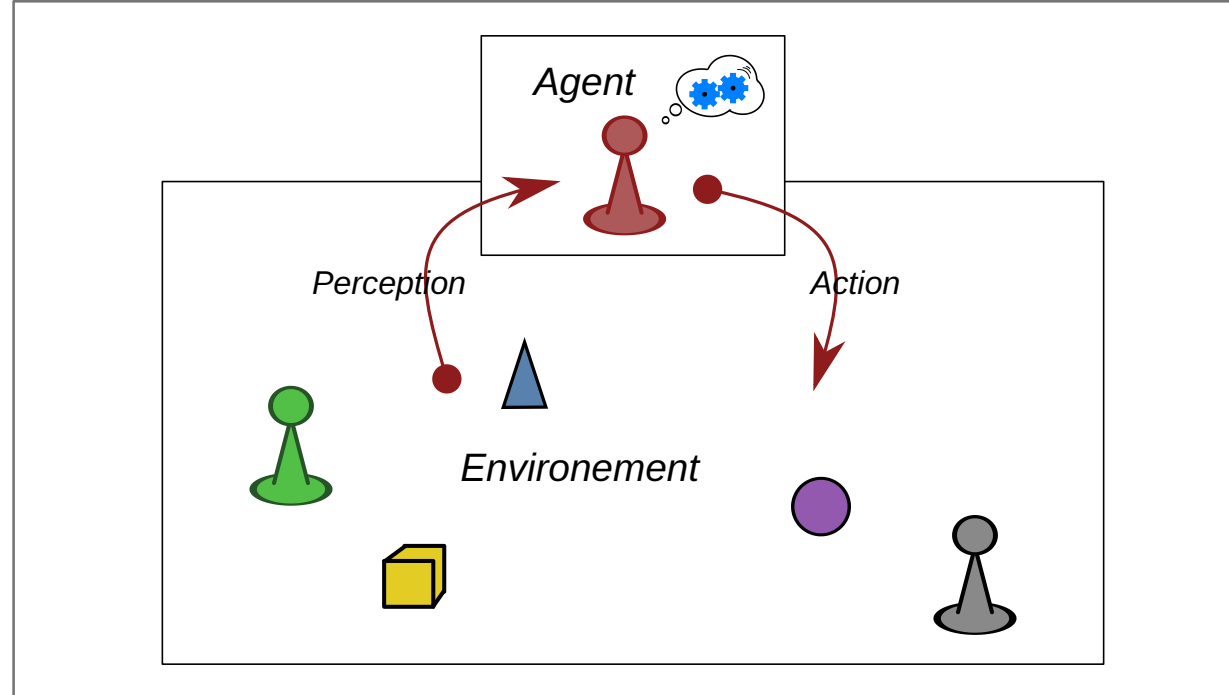


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Acting over a dynamic system: the agent



Rarely deterministic, Mostly uncertain

Rational Agent

"I act, therefore I am."

- ▶ My actions have an effect over the world **AND** I have the choice.
- ▶ *AI*: Model the effect **AND** explore the choices

Deliberativ Architecture - BDI:

- ▶ *Believe*: refers to the knowledge of the agent
- ▶ *Desire*: agent's goals (classically states to reach)
- ▶ *Intention*: succession of actions to perform oriented toward the goals

Acting over a system : formally

Markov Chain (Andreï Markov 1856-1922)

A tuple: $\langle \text{States } (S), \text{Transitions } (T) \rangle$

- ▶ **States:** set of configurations defining the studied system
- ▶ **Transitions:** Describe the possible evolution of the system state

$$T : S \times S \rightarrow [0, 1]$$
$$T(s_t, s_{t+1}) = P(s_{t+1} | s_t)$$

Vocabulary Parrentesis: Hidden Markov Chain

- > The system state is not directly observable.

Acting over a system : formally

Impact of the actions

- ▶ **Actions:** finite set of possible actions to perform

Updated Transition function:

The probabilistic evolution depends on the performed action.

$$T : S \times A \times S \rightarrow [0, 1]$$

$T(s^t, a, s^{t+1})$ return the probability to reach s^{t+1} by doing a from s^t :

$$T(s^t, a, s^{t+1}) = P(s^{t+1} | s^t, a)$$

Multi-Variable Systems

State and Action space:

> Cartesian product over State and Action variables

Multi-variable Transition function:

The probabilistic evolution depends on the performed action.

$$T : S \times A \times S \rightarrow [0, 1] \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, \begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{bmatrix} \right) \in [0, 1]$$

Model of 421: States and actions

► States:

- The value of each die's face ($d_n \in [1, 6]$) and the re-roll number ($h \in [2, 0]$)
- So: **168** states (56 combinations over a horizon of 3).

► Actions:

- The choice of roll again each die: $[roll, keep]$
- so **8** actions (2^3)

Action Example :

By choosing to "roll-*keep*-roll" in state: "6-*4*-3 (2)" to expect a "4-2-1 (1)"

Model of 421: Transition function with 421-game

► Transitions:

- All reachable states by rolling some dice with the probability to reach them.

$$T \left(\begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ h \end{bmatrix}, \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \begin{bmatrix} d'_1 \\ d'_2 \\ d'_3 \\ h' \end{bmatrix} \right) \in [0, 1] \quad \text{example: } T \left(\begin{bmatrix} 6 \\ 5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} r \\ r \\ k \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \\ 1 \\ 1 \end{bmatrix} \right) = 1/36$$

Model of 421: Transition function with 421-game

Transitions Example :

Choosing to "roll-*keep*-roll" from "6-*4*-3 (2)" implies *21* reachable states:

P(...)	=	[0, 1]		P(...)	=	[0, 1]
<i>4</i> -1-1 (1)	=	1/36		...		
<i>4</i> -2-1 (1)	=	1/18		6- <i>4</i> -4	=	1/18
<i>4</i> -2-2 (1)	=	1/36		6-5- <i>4</i>	=	1/18
...				6-6- <i>4</i>	=	1/36

Choosing : building a policy of actions

- ▶ *a policy* (π) : a function returning the action to perform
Considering the current state of the system:

$$\pi : S \rightarrow A$$

$\pi(s)$: the action to perform in s

Choosing : building a policy of actions

Example of policy :

"Always target a 4-2-1": keeping only one **4**, one **2** and one **1**

s	$\pi^{421}(s)$		s	$\pi^{421}(s)$
1-1-1	keep-roll-roll		...	
2-1-1	keep-keep-roll		4-2-1	keep-keep-keep
3-1-1	roll-keep-roll		...	
4-1-1	keep-keep-roll		6-6-5	roll-roll-roll
...			6-6-6	roll-roll-roll

(Invariant over the horizon h)

Automatize Decision Making

Choose an action in a given context (state):

- ▶ Evaluate tuples $\langle s, a \rangle$
- ▶ Selects the best action:

$$\pi^*(a) = \arg \max_{a \in A} (Eval(s, a))$$

Evaluation in Game Theory

Can be done a posteriori, with a lot of data :

- ▶ $Eval(s, a)$ = the probability of winning if doing a in s .

$$Eval(s, a) = P(win \mid \pi(s) = a)$$

- ▶ But: also depends on all the future actions...

Can be done a posteriori, with a lot of *good* data...

Evaluation in 4-2-1

What is the optimal choices ?

- ▶ For example, from 6-1-1 (2): **roll-keep-keep** or **roll-roll-keep** ?

Scores in 421-game

Over the final combination (horizon = 0):

$$\text{score}(4-2-1) = 800$$

$$\text{score}(1-1-1) = 700$$

$$\text{score}(x-1-1) = 400 + x$$

$$\text{score}(x-x-x) = 300 + x$$

$$\text{score}((x+2)-(x+1)-x) = 202 + x$$

$$\text{score}(2-2-1) = 0$$

$$\text{score}(x-x-y) = 100 + x$$

$$\text{score}(y-x-x) = 100 + y$$

Evaluation in 4-2-1

What is the optimal choices ?

- ▶ For example, from 6-1-1 (2): **roll-keep-keep** or **roll-roll-keep** ?

**Requires to evaluate the possible final combinations
after 2 roll-agains...**

- ▶ **It clearly depends on the horizon:**
With an infinite costless roll-agains
it is always preferable to target the **4-2-1**

Exercices, compute a policy

Basic learning process :

- ▶ Record a player experience (trace)
- ▶ Compute average values for tuples $\langle s, a \rangle$
- ▶ Use those values to select the actions.

But: depends on the initial player behavior...

Make it iterative :

- ▶ Use your new player to create new data *and* learn again...

(AlphaGo Zero 2017 wins 100/0 AlphaGo Lee)